# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: Artificial Intelligence COURSE CODE: CSA17**

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**COURSE OUTCOME COVERED**

***CO2:*** *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

*Assessment Tool Weightage: 40%*

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**QUESTIONS: 3**

**Duration: 1 Hour**
**Total Marks:30**

Annexure A

**sDry Run Evaluation**

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***Code of Conduct:***

*I Arava Raja(192365033) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action*

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***Arava Raja***
***Signature of the student***

Annexure A

**Dry Run Evaluation**

**1.** Question 1 – Breadth-First Search (BFS)

Consider the following graph:

```
        A
       / \
      B   C
     /\   /\
    D  E F  G
```

**Task:** Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

| **Iteration** | **Queue** | **Visited Node** |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| 7 | | |

**Questions:**

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

```
        A
       / \
      B   C
     /\   /\
    D  E F  G
```

| Iteration | Stack | Visited Node |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| 7 | | |

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**

   **1 3 6**
   **5 0 2**
   **4 7 8**

**Goal State :**

**1 2 3**
**4 5 6**
**7 8 0**

## Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

| Iteration | Current State | Queue Before Expansion | Queue After Expansion |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |

## Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

**RUBRICS FOR CALCULATION**

| Criteria | Weightage (%) | Level 4 – Exemplary (4) | Level 3 – Proficient (3) | Level 2 – Developing (2) | Level 1 – Beginning (1) |
|---|---|---|---|---|---|
| **Understanding of Search Problem** | **20** | Clearly understands the search problem, initial state, goal state, and search strategy. | Understands the problem with minor misunderstandings. | Demonstrates partial understanding of the search problem. | Shows little or no understanding of the search problem. |
| **State Expansion and Dry Run Execution** | **30** | Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors. | Performs most iterations correctly with minor mistakes in state expansion or data structure updates. | Performs only some iterations correctly; several errors in tracing or node expansion. | Unable to correctly perform the dry run or expand states. |
| **Application of Search Algorithm** | **20** | Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules. | Applies the algorithm correctly with a few procedural errors. | Applies only basic algorithm steps with noticeable mistakes. | Fails to apply the correct search algorithm. |
| **Accuracy of Solution Path** | **20** | Identifies the correct traversal order/solution path and goal state with proper justification. | Solution path is mostly correct with minor errors. | Partially correct solution path; goal may not be reached correctly. | Incorrect or incomplete solution path. |
| **Presentation and Logical Reasoning** | **10** | Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented. | Presentation is clear with minor organizational issues. | Limited explanation and organization; reasoning is partially clear. | Poor presentation with unclear or missing reasoning. |

## Q1: Breadth-First Search (BFS)

### Given Graph

```
      A
    /   \
   B     C
  / \   / \
 D   E F   G
```

### Dry Run Table:

| Iteration | Queue Before | Visited Node | Queue After |
|---|---|---|---|
| 1 | A | A | B, C |
| 2 | B, C | B | C, D, E |
| 3 | C, D, E | C | D, E, F, G |
| 4 | D, E, F, G | D | E, F, G |
| 5 | E, F, G | E | F, G |
| 6 | F, G | F | G |
| 7 | G | G | Empty |

### BFS Traversal Order

A → B → C → D → E → F → G

### Queue After Every Iteration

| Iteration | Queue |
|---|---|
| 1 | B, C |
| 2 | C, D, E |
| 3 | D, E, F, G |
| 4 | E, F, G |
| 5 | F, G |
| 6 | G |
| 7 | Empty |

**Q 2: Depth-First Search (DFS)**
Using the same graph.

**DFS Traversal**
Starting from **A**, always visit the left child first.
Traversal:
A → B → D → E → C → F → G

**Dry Run Table (Stack Concept)**

| Iteration | Stack Before | Visited Node | Stack After |
|---|---|---|---|
| 1 | A | A | C, B |
| 2 | C, B | B | C, E, D |
| 3 | C, E, D | D | C, E |
| 4 | C, E | E | C |
| 5 | C | C | G, F |
| 6 | G, F | F | G |
| 7 | G | G | Empty |

**DFS Traversal Order**
A → B → D → E → C → F → G

**Q 3: 8-Puzzle BFS**
**Initial State**
1 3 6
5 0 2
4 7 8
Goal State
1 2 3
4 5 6
7 8 0

**Iteration Table (Conceptual BFS Dry Run)**

| Iteration | Current State | Queue Before Expansion | Queue After Expansion |
|---|---|---|---|
| 1 | Initial State | Initial | Generate all valid moves (Up, Down, Left, Right) |
| 2 | First generated state | Remaining states | Expand next state |
| 3 | Next state | Updated Queue | Add newly generated states |
| 4 | Next state | Updated Queue | Continue expansion |
| 5 | Goal State Found | Updated Queue | Search Stops |

**Answers**

**1. Which node/state is expanded in each iteration?**

- Iteration 1 → Initial State
- Iteration 2 → First generated state
- Iteration 3 → Second generated state
- Iteration 4 → Third generated state
- Iteration 5 → Goal State

**2. How many states are generated?**

The exact number **cannot be determined from the assessment sheet alone**, because the intermediate BFS expansion order (move priority such as Up/Down/Left/Right and duplicate-state handling) is not specified.

**3. Solution Path**
Initial State
↓

Move Blank

↓

Intermediate State

↓

Intermediate State

↓

Goal State
(The exact sequence depends on the BFS expansion order, which is not provided in the question.)

**4. Why does BFS always find the shortest solution?**
**Answer:**
Breadth-First Search explores all states level by level. Since every move in the 8-puzzle has the same cost, BFS reaches the goal using the minimum number of moves. Therefore, BFS always guarantees the shortest solution path in an unweighted search problem.

**Final Answers to Write**
**BFS**
Traversal:
A → B → C → D → E → F → G
**DFS**
Traversal:

A → B → D → E → C → F → G

**Why BFS is Shortest?**

BFS explores nodes level by level.
Since every move has equal cost,
the first time the goal is found,
it is guaranteed to be the shortest path.